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A tectonically driven 60 million-year biogeochemical redox cycle paces marine biodiversity

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The fossil record shows a prominent 60 million-year biodiversity cycle during the Phanerozoic Eon, the origin of which is still unknown. Here we use time-series analysis and correlation of empirical and model datasets of Earth's interior and surficial processes to demonstrate that this cycle is a pervasive feature in marine animal genus-level diversity data that dominates in the Paleozoic Era. Our results suggest that extinctions are likely the primary driver of this observed cyclicity. We detected a correlatable 60 million-year cyclicity in global tectonics, and in marine $^{87}\text{Sr}/^{86}\text{Sr}$ and $\delta^{34}\text{S}$ isotopes, all of which are dominant in the Paleozoic. We conclude that continental weathering driven by global tectonic degassing and building of continental arcs may have in turn controlled paleo-seawater redox cycling during the Paleozoic when oceans were likely less saturated with respect to oxygen. In particular, we suggest that the 60 million-year fluctuations in biotic diversity are responses of shallow marine habitats to the combined effects of continental weathering and redox cycling, under global tectonic control.

There is increasing awareness of the critical role of ocean redox state (i.e., high vs low dissolved oxygen levels in seawaters) in driving the Phanerozoic variations in marine faunal diversity and major biological crises and mass extinctions^{1–7}. A long-term Phanerozoic decline, from Cambrian to present day, is apparent in extinction and origination rates in the marine animal fossil record, based on a broad range of rate metrics^{8–10}. This trend has recently been linked to long-term changes in the Neoproterozoic and Paleozoic climate and ocean oxygenation⁶. In addition to these long-term evolutionary and environmental trends in Earth's history, geochemical modeling results also provide evidence for variations in marine redox conditions and thus dissolved-oxygen concentrations in paleo-seawater throughout the Phanerozoic^{7,11,12}. Additionally, observations of present-day climate and ocean dynamics show that variations in dissolved oxygen concentrations and global warming exert a strong control on marine biodiversity^{13–15}. These modern-day biotic sensitivities can potentially inform our ideas and better understanding of the drivers of past extinctions occurring in response to climate and other environmental changes¹⁶.

Phanerozoic marine faunal genus diversity data show cyclic variations with periodicities of few to several tens of million years (Myr) superimposed on long-term evolutionary trends^{17,18}. By removing the increasing trend in biodiversity from Cambrian to present day, a prominent and well-defined

cyclicity of ~60 Myr, representing a 20% of the residual variance, emerges in the genus biodiversity data^{17,18}. This ~60 Myr cyclicity accounts for ~13% of the residual variance during the Meso-Cenozoic interval and up to 22% during the Paleozoic Era. The prominence of such ~60 Myr cycle in the fossil record, which maps closely the boundaries of geological Periods¹⁸, is consistent with the fact that many geological period boundaries were originally recognized and defined on the basis of recognizable biotic turnovers^{19,20}. Hence, geological Period boundaries are not arbitrary subdivisions of time but rather reflect major qualitative biotic changes or turnovers in Earth system functions and ocean chemistry relevant to the macro-evolution of marine life¹⁸. However, thus far, the underlying driving mechanism(s) of this 60 Myr marine biodiversity cycle remain unexplained¹⁷. Here, we explore the causative link(s) between marine redox conditions and continental weathering processes (recorded by marine $^{87}\text{Sr}/^{86}\text{Sr}$ and $\delta^{34}\text{S}$ proxies, respectively), which in turn are controlled by changes in global plate kinematics⁷ and fluctuations in the seawater oxygen levels, which we link to the diversity of marine animal life during the Phanerozoic.

We use empirical and model datasets of Earth's interior and surficial processes, including recent global tectonic plate reconstructions, marine generic biodiversity from the Paleobiology Database (PBDB, see Data and

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Methods), and biogeochemical proxy records of ocean redox ($\delta^{34}\text{S}$) and continental weathering ($^{87}\text{Sr}/^{86}\text{Sr}$) conditions. These various datasets are used to assess their potential common and correlatable ~ 60 Myr cyclicality to constrain and better understand their causal links and identify the primary drivers of such observed cyclicality. We have also explored the unresolved issue of the preservational/sampling bias on the record of secular variations of biodiversity and other related geological and isotope proxies^{21–25}.

Results

The 60 Myr fossil and sedimentary cycles

Spectral analysis of *range-through* (RT) and *Shareholder Quorum Sub-sampling* (SQS) marine faunal genus-level biodiversity datasets over the Phanerozoic Eon (PBDB, see Data and Methods) shows a pronounced, common cycle band of ~ 60 to ~ 61 Myr (Fig. 1 and Supplementary Fig. S1). Spectral analysis of marine genus-level extinction rates also shows a 60 Myr cycle band ranging from 61 to 63 Myr (Fig. 1). The 60 Myr biodiversity and extinction cycles are both characterized by stronger powers during the Paleozoic (Fig. 2 and Supplementary Figs. S2 and S3). Spectral analysis of Phanerozoic macrostratigraphic data of total marine carbonate and sediment area in North America (see Data and Methods) reveals two prominent peaks of 67.5 and 56.3 Myr in total sediment, and 67.5 and 50.9 Myr in

carbonate that may average ~ 62 Myr and ~ 59 Myr signals respectively (Fig. 3 and Supplementary Fig. S4). Thus, we have focused on the amplitude of the 60 Myr carbonate cycle rather than its periodicity (Fig. 3, see Discussion). The global nature of the 60 Myr North America macrostratigraphic cycle has been explored on the basis of spectral analysis of non-North American PBDB fossil data vs North America PBDB fossil data. Results point to a common, correlatable 60 Myr biodiversity cycle (Supplementary Fig. S5), supporting the hypothesis that North America data share global components of variability²⁶ (Data and Methods). Finally, spectral analysis of the variations of duration of the standard geological Stages shows a strong peak of ~ 60 Myr (Supplementary Fig. S4). The ~ 60 Myr biodiversity cycle tracks the duration of the global geological periods, with their minima being in phase with most period boundaries (Fig. 1). There are two exceptions, the Ordovician and Silurian Periods, which together comprise a 60 Myr oscillation; the second exception is for the Cenozoic Era, which is another 60 Myr oscillation. Additionally, our analyses provide several arguments against the suggestion that the 60 Myr biodiversity cycle arises solely from a preservational/sampling bias in fossil data, based on the following observations: (i) the extracted 60 Myr cyclicality from variations of the duration of the geological stages does not correlate with most of boundaries of geological Periods (Supplementary Fig. S4), (ii)

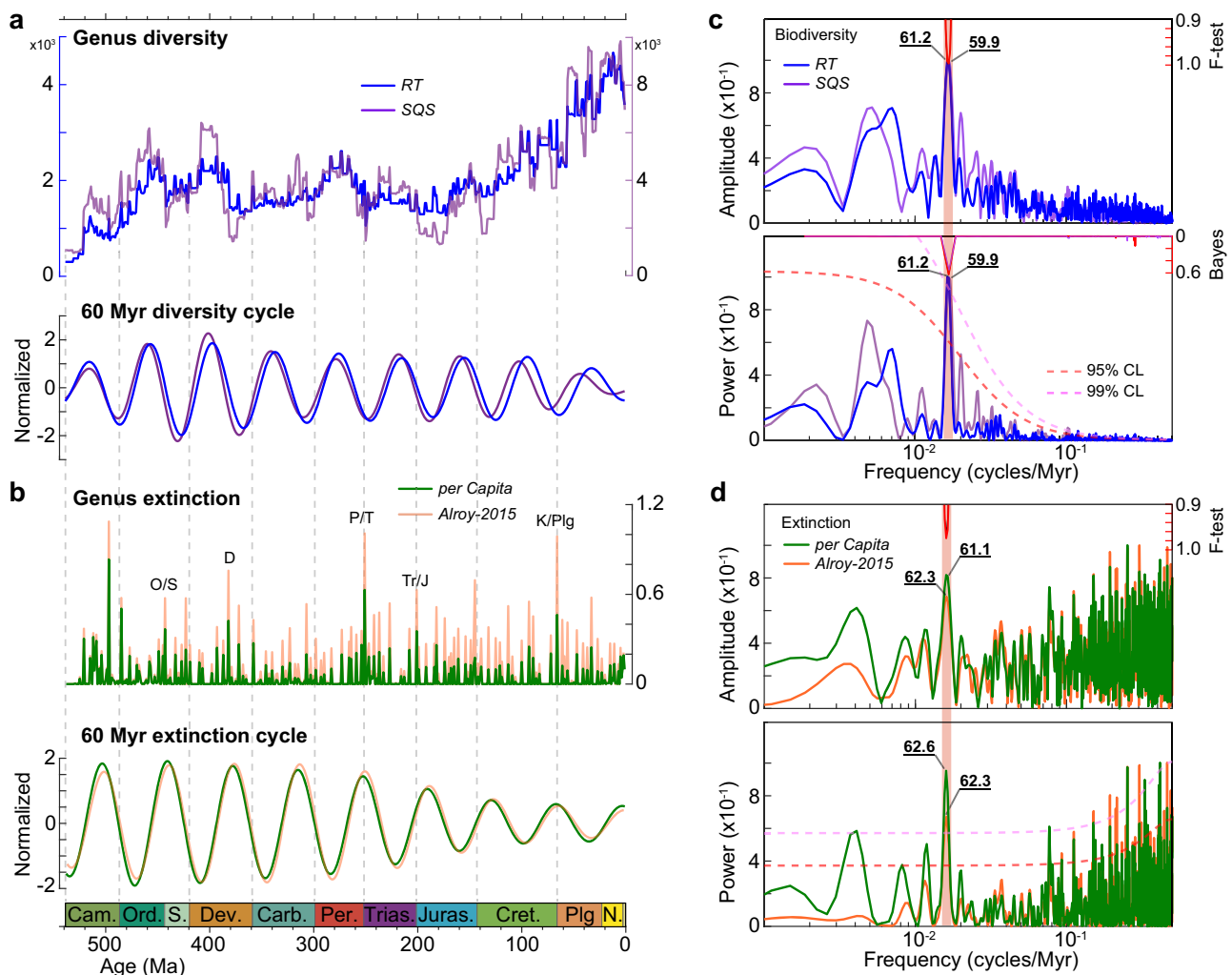
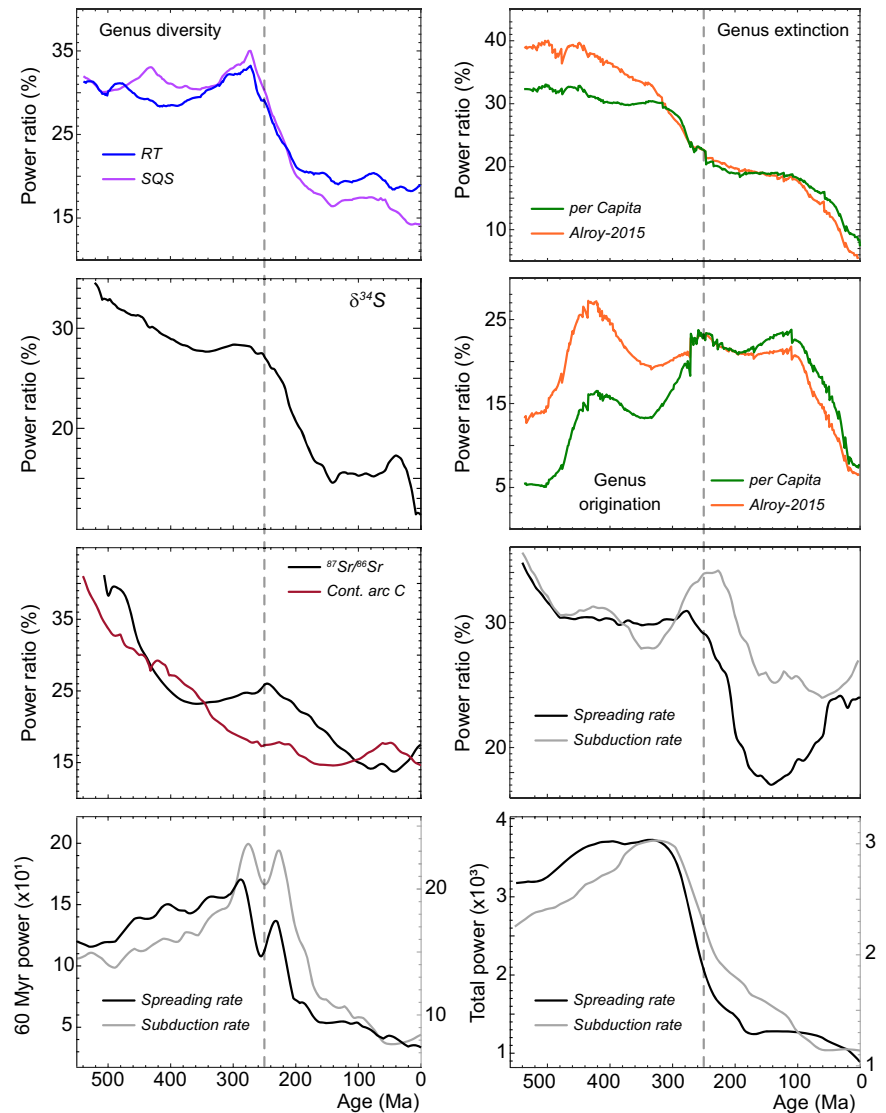


Fig. 1 | Time-series analysis of global marine fossil data. a Raw genus-level diversity data using two metrics (*range-through*: RT, and *Shareholder Quorum Sub-sampling*: SQS, see Data and Methods), along with the extracted 60 Myr cycle. **b** Raw genus-level extinction data using two metrics (*per Capita* and *Alroy-2015*, see Data and Methods), along with the extracted 60 Myr cycle. The five major mass extinctions are shown (O/S, end-Ordovician; D, late-Devonian; P/T, end-Permian; Tr/J, end-Triassic; K/Pg, end-Cretaceous).

c, d Spectral analysis of the respective biodiversity and extinction data shown in 'a' and 'b' (same color codes). Prior to spectral analysis, the SQS data were 50% weighted-average detrended, while the RT data were 35% weighted-average detrended (see Supplementary Fig. S1 for details). Significance tests are for SQS diversity and *per Capita* extinction data. Significance is higher in RT diversity and in *per Capita* extinction data.

Fig. 2 | Evolution of the power of the 60 Myr cycle band recorded in diversity, extinction, origination, biogeochemical $\delta^{34}\text{S}$ and $^{87}\text{Sr}/^{86}\text{Sr}$ proxy datasets, and tectonic data during the Phanerozoic. The power ratio is expressed as the power of the 60 Myr cycle band (1/70 to 1/50 cycles/Myr) over the power of the total frequency bands (see Data and Methods). The power of the 60 Myr cycle band, and the total power of cycles from 0 to 0.1 cycles/Myr are shown for the tectonic data (bottom, left and right panels).



the 60 Myr cyclicities in biodiversity and macrostratigraphic sediment data are incoherent (coherence close to zero, Supplementary Figs. S6 and S7), and (iii) the 60 Myr cycle in the number of fossil global occurrences, a more direct measure of fossil sampling intensity, does not correlate with the 60 Myr biodiversity cycle over the Phanerozoic (Supplementary Fig. S7), although some coherent phase relationship can be observed in the Cambrian through the Carboniferous where global occurrences led biodiversity (Supplementary Fig. S6). This may support the observed and statistically significant correlation among biodiversity, global occurrences and sediment coverages during the Paleozoic²⁷. The detection of the 60 Myr cycle in other geological datasets that were resolved independently of the consideration of geological boundaries of stages and periods (Fig. 2) further supports a common-cause mechanism and weakens the sampling-bias hypothesis (see also the discussion of sampling bias in ref. 18).

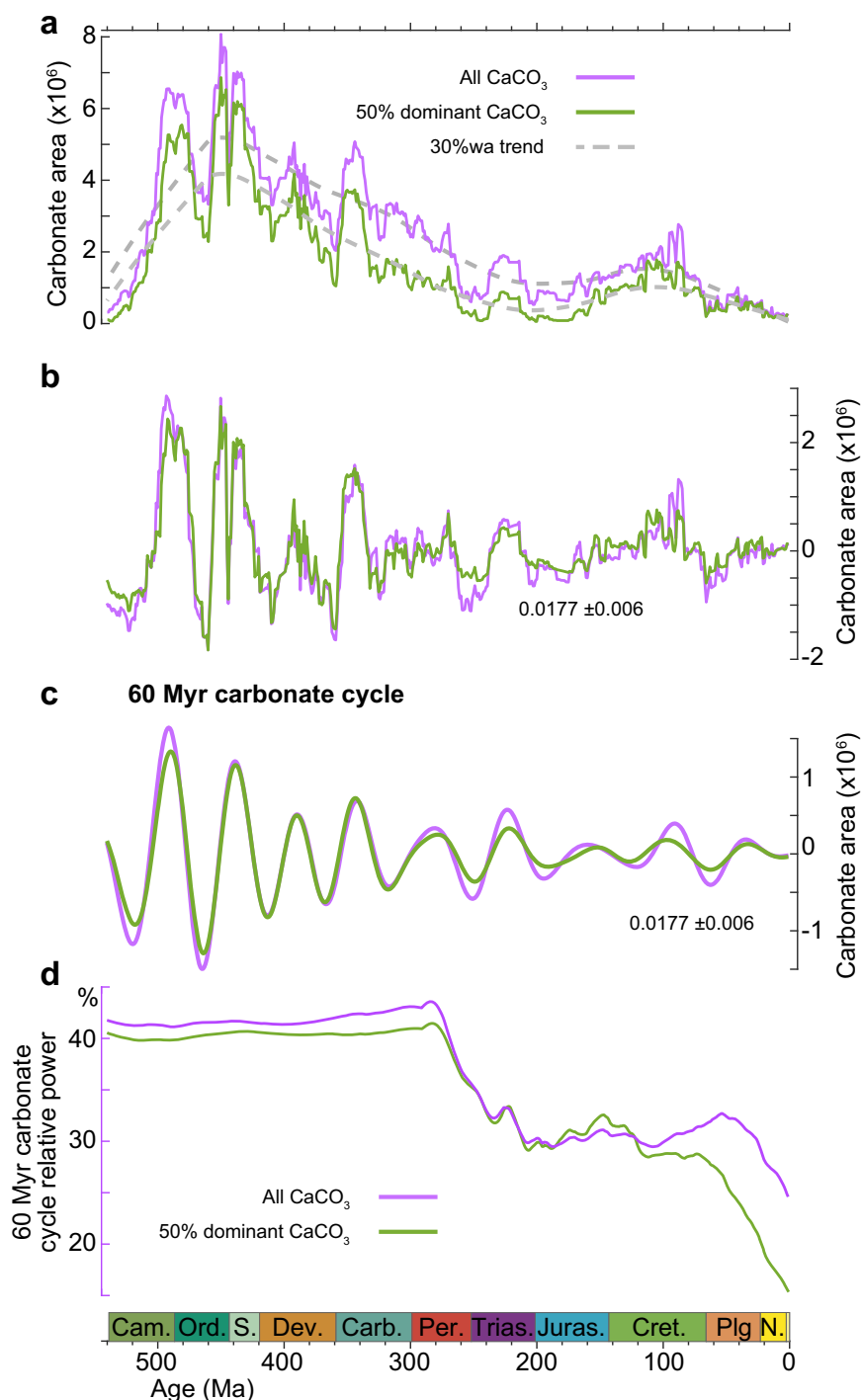
The observed 60 Myr biodiversity signal arises from both extinction and origination rates but with an apparently stronger contribution from the extinction (Figs. 1 and 2), which has higher fidelity in the face-value fossil record²⁸. The power of the 60 Myr origination cycle spreads over most of the Phanerozoic, with two inflection points, one at the Silurian and the other at the mid-Cretaceous, while the power of the 60 Myr extinction cycle gradually decreases towards recent times with an accelerated decline starting from the early Eocene (Fig. 2). The 60 Myr biodiversity cycle shares the same long-term behavior as the extinction cycle and shows much more power

focused on the Paleozoic, with a notable decrease towards recent times (Fig. 2). Finally, the 60 Myr biodiversity and extinction cycles are in anti-phase, with extinction leading biodiversity (Supplementary Fig. S8 and Table S1). This further supports that extinction mechanisms, rather than origination, are the primary driver of observed cyclicity. Although origination rates have a weak and non-significant 60 Myr signal (Supplementary Fig. S3), they should have a significant contribution to biodiversity variations given the longer phase lag (12.1 Myr) of biodiversity with respect to extinction (Supplementary Table S1).

The 60 Myr marine biogeochemical and tectonic cycles

Spectral analysis of the radiogenic Sr isotope ($^{87}\text{Sr}/^{86}\text{Sr}$) record of the Phanerozoic marine carbonates shows a prominent 58.5–60.8 Myr cycle (Fig. 4), exhibiting more power during the Paleozoic Era (Fig. 2). Spectral analysis of the Phanerozoic marine sulfate $\delta^{34}\text{S}$ record shows a similar and prominent 60.2–65.6 Myr cycle (Fig. 4 and Supplementary Fig. S9), also exhibiting more power during the Paleozoic Era (Fig. 2). In order to explore the possible drivers or causative links of such cyclicity in the oceanic Sr and S cycles and their isotope records, including phenomena linked to the Earth's interior dynamics such as mid-ocean ridge-related processes²⁹, we performed spectral analysis on the modeled global seafloor spreading and subduction rates spanning the Phanerozoic (see Data and Methods). Results show a common and pronounced peak of ~37 to

Fig. 3 | Time-series analysis of macrostratigraphic carbonate data (see Data and Methods). a Raw data along with long-term weighted average (wa) trends. **b** Detrended data. **c** The 60 Myr bandpass (indicated in cycles/Myr) filtered data. **d** Evolution of the power of the 60 Myr carbonate cycle band over the Phanerozoic. The power ratio is expressed as the power of the 60 Myr cycle band (1/70 to 1/50 cycles/Myr) over the power of the total frequency bands (see Data and Methods) in the detrended data in 'b'. Extended analyses of macrostratigraphic carbonate data are provided in Supplementary Fig. S4.



~38 Myr cyclicity in spreading and subduction rates, which has already been detected for the Ceno-Mesozoic interval¹⁸, but notably also another common and significant ~59 to ~60 Myr spectral peak (Fig. 5 and Supplementary Fig. S10). The latter is thus identical or very similar to a cycle band detected in the marine $^{87}\text{Sr}/^{86}\text{Sr}$ and $\delta^{34}\text{S}$ proxy records (Fig. 4). In addition, the observed ~60 Myr cyclicities in spreading and subduction rates are in phase, pointing to a potential coupling between mid-ocean ridge and subduction related processes, and thus their possible impact on ocean chemistry and the evolution of the marine biota (see Discussion). Finally, the spectral analysis of carbon (C) flux into the atmosphere-ocean systems derived from global continental arcs also shows a ~63 Myr cycle (Fig. 5 and Supplementary Fig. S15).

Overall, the observed 60 Myr cycle bands in marine $^{87}\text{Sr}/^{86}\text{Sr}$ and $\delta^{34}\text{S}$ records, tectonic data, and biodiversity point to a common driver and causative link between Earth's interior dynamics, Phanerozoic seawater chemistry, and marine biodiversity mediated via changes in continental weathering, oceanic redox cycling and related environmental/biological changes. We thus compared, at the 60 Myr cycle band, tectonic vs marine biogeochemical datasets on the one hand, and tectonic vs biodiversity datasets on the other hand, to look for potential connections between them (Supplementary Fig. S8 and Table S1). We found a strong antiphase relationship between spreading rates and biodiversity, while spreading rates also show an in-phase relationship with marine sulfate $\delta^{34}\text{S}$ over the Phanerozoic. Notably, the latter proxy ($\delta^{34}\text{S}$) shows a striking antiphase relationship

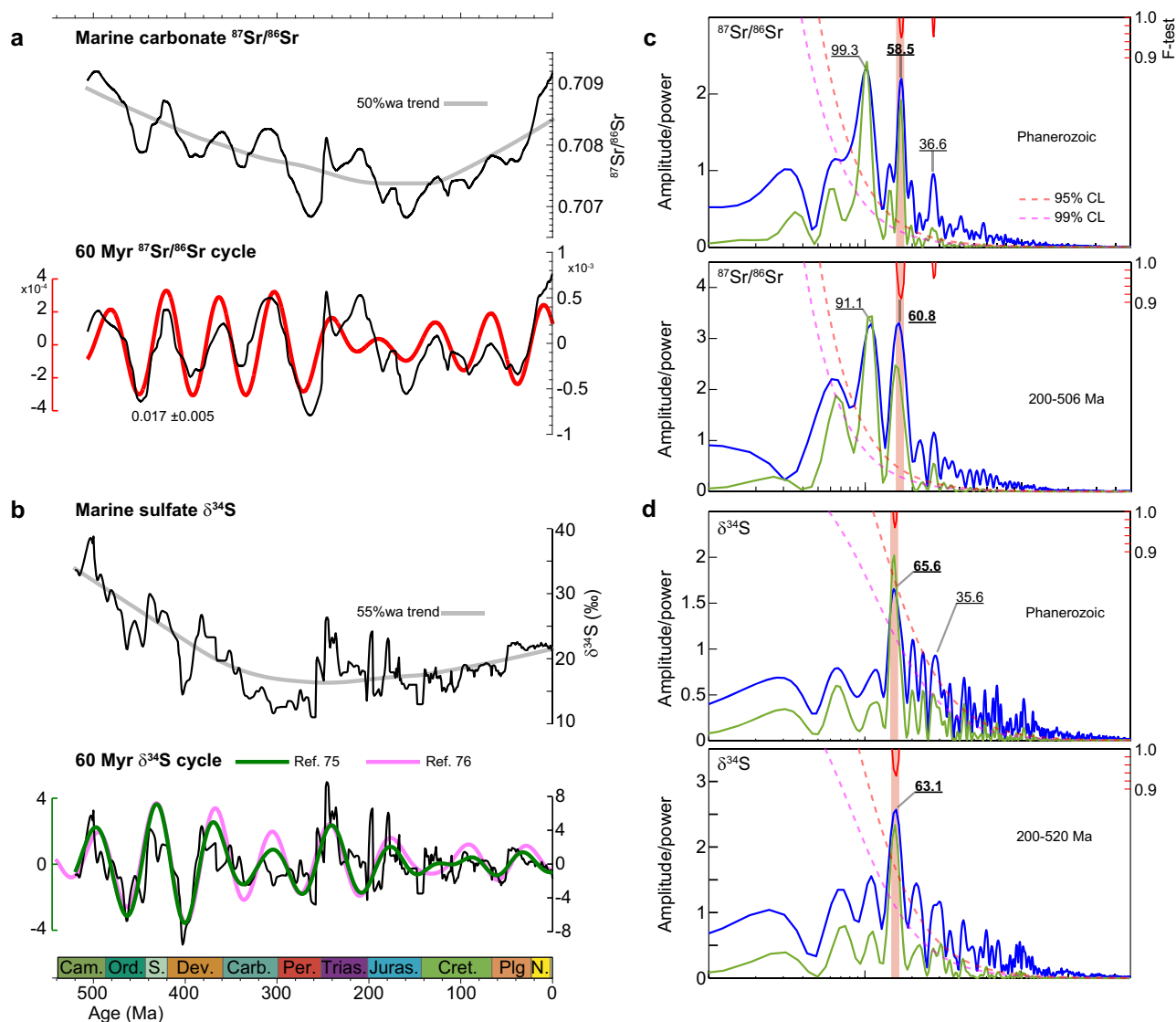


Fig. 4 | Time-series analysis of biogeochemical datasets. **a** Upper curves: raw marine carbonate strontium isotope ratios ($^{87}\text{Sr}/^{86}\text{Sr}$) and weighted average (wa) fit (see Data and Methods). Lower curves: Detrended data, along with 60 Myr bandpass filtered data. **b** Upper curves: raw marine sulfate sulfur isotopes ($\delta^{34}\text{S}$) and wa fit (see Data and Methods). Lower curves: detrended data and 60 Myr bandpass filtered

data. Extended data and analyses of $\delta^{34}\text{S}$ record are provided in Supplementary Fig. S9. **c, d** 2π -MTM amplitude spectra (blue) versus periodograms (green) (see Data and Methods) of the data shown in 'a' and 'b'. The orange-shaded vertical bar depicts the spectral peaks at the 60 Myr cycle band.

with marine biodiversity, which is most evident during the Paleozoic. Additionally, there is a strong coherence among extinction and marine $\delta^{34}\text{S}$, with the latter leading or pre-dating extinction events (Supplementary Fig. S8) although the residence time of sulfate is long (ca. 10 Myr)³⁰. Marine $^{87}\text{Sr}/^{86}\text{Sr}$ and $\delta^{34}\text{S}$ are also in phase, where the $^{87}\text{Sr}/^{86}\text{Sr}$ signal is somewhat (~9 Myr) ahead of the phase. Finally, the 60 Myr cycle bands in continental arc C outflux and marine $^{87}\text{Sr}/^{86}\text{Sr}$ are also in phase during the Paleozoic (Supplementary Fig. S15) when the 60 Myr cycle powers are the strongest (Fig. 2). These results seem to support a hypothesis invoking the substantial role of continental arc CO_2 degassing driving continental weathering processes³¹, which in turn modulated nutrient inputs into the oceans, thereby affecting marine biodiversity and oceanic biogeochemical cycles, including marine paleo-redox conditions.

Discussion

Tectonically driven 60 Myr marine biogeochemical cycles

The in-phase relationship between the 60 Myr seafloor spreading and subduction rates and the marine sulfate ($\delta^{34}\text{S}$) cycling suggests a potential causal link between the redox-driven S isotopes (via sulphate-sulphide

speciation, see below) in the Phanerozoic ocean and global tectonics. The oceanic sulfate $\delta^{34}\text{S}$ record through time provides an estimate of temporal changes in dissolved S species present in the ocean, namely the oxidized form (i.e., sulfates or dissolved SO_4^{2-} ions) and the reduced form (sulfides or H_2S , HS^-), the latter leading to the formation and deposition of marine pyrite (FeS_2)³². Accordingly, higher or more positive marine $\delta^{34}\text{S}$ would suggest an increase in sulfide/pyrite precipitation and burial under more reducing and oxygen-poor marine redox conditions³³, and vice versa, since the expansion of sulfidic or anoxic conditions in the ocean depends on marine paleo-redox state^{33,34}.

One long-accepted hypothesis is that greater mid-ocean ridge spreading and accretion rates would induce a substantial rise of sea level at this timescale, slowing down oceanic circulation and promoting the onset of anoxic or O_2 -depleted paleo-seawater redox conditions^{3,35,36}. However, the Phanerozoic sea-level record does not exhibit a 60 Myr cycle¹⁸, calling for a different driver linking the global tectonics to changes in the Phanerozoic seawater $\delta^{34}\text{S}$ record and thus marine paleo-redox conditions. Here we propose two potential mechanisms for the above observations: (i) the release of reduced gases and fluids from submarine volcanism into the oceans,

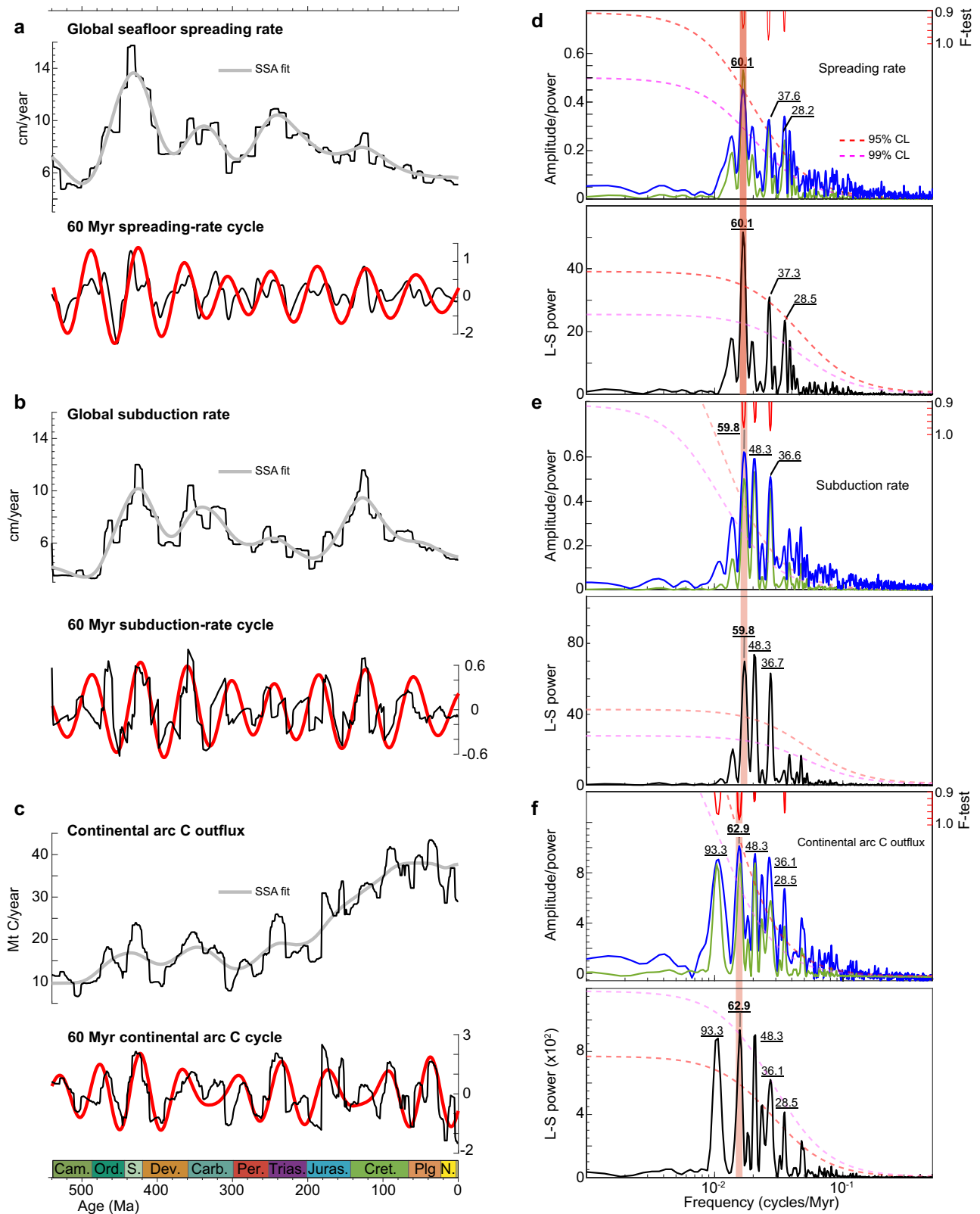
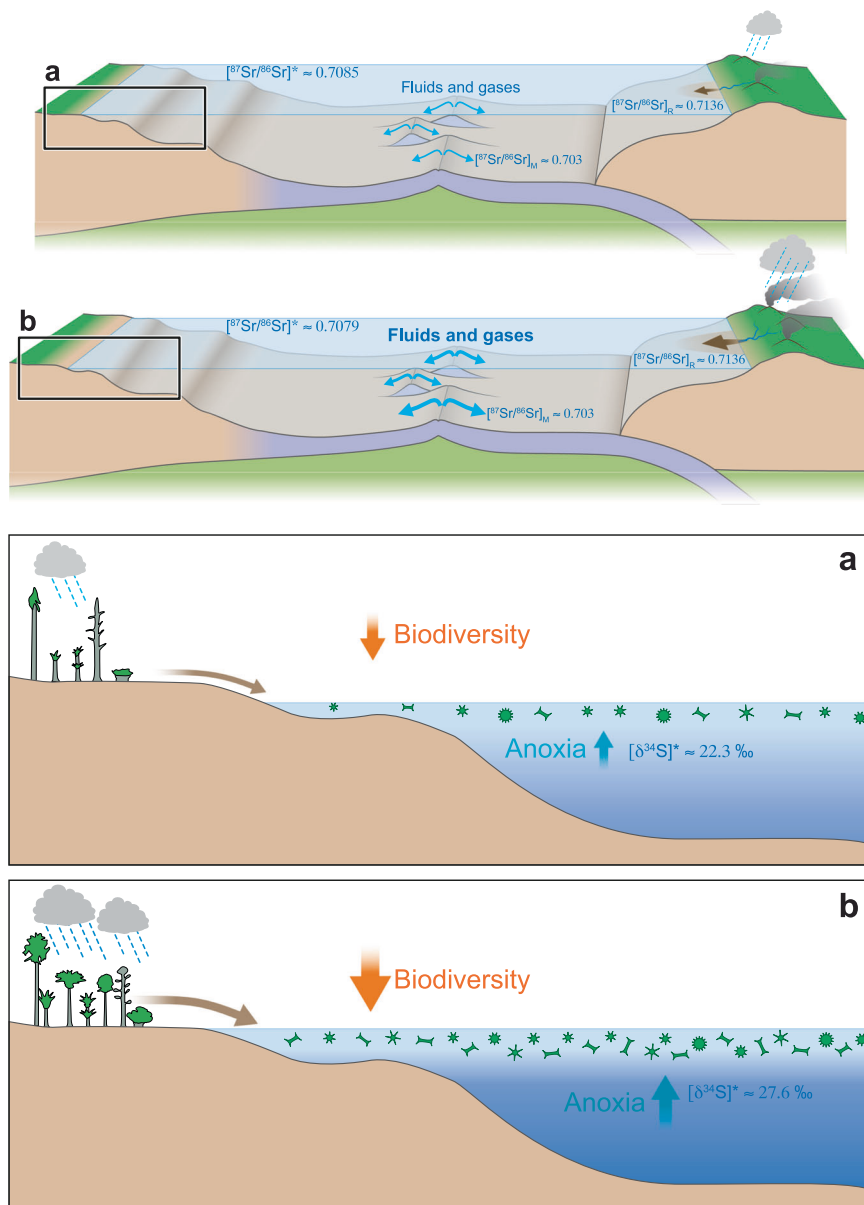


Fig. 5 | Time-series analysis of tectonic datasets. **a** Upper curves: raw mid-ocean spreading rates (see Data and Methods), along with singular spectrum analysis (SSA) low-frequency reconstructed components (gray curve, embedding dimension $M = 100$ Myr). Lower curves: residuals of SSA fit, along with the 60 Myr bandpass filtered data. **b** The same caption as in 'a' but for global subduction rates. Extended analyses of tectonic data are provided in Supplementary Figs. S10 and S15. **c** The

same caption as in 'a' but for continental arc C outflux data (see Data and Methods). **d, e, f** 2π -MTM amplitude spectra (blue) versus periodograms (green) and Lomb-Scargle (L-S) periodogram (black) (see Data and Methods) of the data shown in 'a', 'b' and 'c'. The orange-shaded vertical bar depicts the spectral peaks at the 60 Myr cycle band.

Fig. 6 | Schematic scenarios illustrating possible driving mechanisms of the 60 Myr biodiversity cycle during the Phanerozoic. **a** Relatively stronger global tectonic activity, recorded in spreading and subduction rates (Fig. 5), favors the release of reduced fluids and gases at MORs, CO_2 degassing and mountain building (island and continent arcs) stimulating the weathering process expressed in marine carbonate $^{87}\text{Sr}/^{86}\text{Sr}$ proxy (Fig. 4), nutrient and alkalinity supplies into the ocean, favoring seawater fertilization and marine productivity, hence oxygen overconsumption, and the consequent expansion of anoxia and euxinia expressed in $\delta^{34}\text{S}$ proxy (Fig. 4). This would increase extinction rates and decrease biodiversity (Fig. 1). **b** The opposite scenario. The size of the arrows indicates the relative intensity of the process. $^{87}\text{Sr}/^{86}\text{Sr}_R$ and $^{87}\text{Sr}/^{86}\text{Sr}_M$ values of riverine and mantle components respectively are from Allègre et al.⁸⁶ $^{87}\text{Sr}/^{86}\text{Sr}^*$ and $[\delta^{34}\text{S}]^*$ values are extracted from the proxy records as maximal and minimal amplitudes at the 60 Myr cycle band.



which in turn would generate and maintain more reducing marine paleoredox conditions^{3,37} (Fig. 6), and (ii) tectonically controlled degassing of volcanogenic CO_2 that would modulate the global weathering rates, as recorded in marine $^{87}\text{Sr}/^{86}\text{Sr}$ ratios (Figs. 4 and 5). The latter process would also supply higher input of nutrients into the ocean or epeiric seas, further promoting marine bioproductivity and thus organic carbon production and burial in the ocean. These biologically-driven processes would in turn have impact on marine paleoredox conditions via oxidation of organic carbon that consumes O_2 , and thus on the marine $\delta^{34}\text{S}$ proxy record (Figs. 4 and 6).

Submarine volcanism and hydrothermal activity at mid-ocean ridges (MORs) can inject massive amounts of magma-derived volatiles and fluids into the ocean, which are rich in reduced chemical compounds, such as reduced sulfur and carbon species: H_2 , H_2S and CH_4 ^{38,39}, and reduced metals such as Fe(II) ⁴⁰. These volcanic and reduced gases and metals – acting as electron donors – are then also readily oxidized in contact with seawater^{37,41} either via reduction of marine sulfate (SO_4^{2-}) or via other oxidized species/electron acceptors in seawater, thus causing progressive oxygen depletion and eventually anoxic/euxinic marine redox conditions^{37,41–44}. In addition, increased sulfate reduction in the ocean facilitated via reduced volcanic gases/metals or alternatively via marine organic matter (carbon), acting as

electron donor, would also produce an enhanced carbonate alkalinity flux (input of CO_3^{2-} and HCO_3^- species) into the ocean, thus causing seawater CaCO_3 oversaturation, the latter associated with increased authigenic and redox-driven carbonate precipitation and deposition in the ocean^{45,46}. This redox-driven ‘alkalinity overshoot’ process is predominately expected to lead to an enhanced authigenic carbonate formation at a seafloor or the sediment/seawater interface⁴⁵, and might partly explain the observed coupling between marine carbonate (CaCO_3) coverage/area at continental shelves and paleo-seawater sulfur isotope ($\delta^{34}\text{S}$) records, and their common ~60 Myr cyclicity (Figs. 2 and 3). An alternative and more dominant source of alkalinity to the oceans should be derived from continental weathering⁴⁶, as reflected in the common ~60 cyclicity in the marine $^{87}\text{Sr}/^{86}\text{Sr}$ record and marine carbonate coverage/area (Figs. 3 and 4).

Additionally, our results show that faster spreading rates correlate with increased (more radiogenic) marine $^{87}\text{Sr}/^{86}\text{Sr}$ ratios, implying that the main driver of the 60 Myr cyclicity in the oceanic Sr isotope budget is not MOR basalt production, as the latter should decrease $^{87}\text{Sr}/^{86}\text{Sr}$ in seawaters. Thus we propose that increases in marine $^{87}\text{Sr}/^{86}\text{Sr}$ predominantly arise from enhanced continental weathering fluxes to the ocean, as typical continental crust or felsic rocks (granite, granodiorite) are significantly enriched in the

radiogenic Sr (i.e., having higher $^{87}\text{Sr}/^{86}\text{Sr}$) compared to the mantle, mafic rocks or oceanic crust. Our results also demonstrate that the 60 Myr cycle in marine $^{87}\text{Sr}/^{86}\text{Sr}$ is in phase with the 60 Myr global subduction rate cycle (Supplementary Fig. S8), hinting at a potential connection between the continental weathering flux of radiogenic Sr into the ocean originating from differential uplift of crustal terrains or continental arcs driven by changing convergence rates (Fig. 5 and Supplementary Fig. S14). Hence, we speculate that continental weathering fluxes are modulated at longer (~60 Myr) timescales by the intensity of mountain building and volcanogenic CO_2 degassing along ocean margins driven by plate convergence, mediated via subduction processes (Fig. 6). An extended discussion on the application of marine $^{87}\text{Sr}/^{86}\text{Sr}$ ratios as a weathering proxy is provided in the 'Data and Methods' Section.

The hypothesis of higher marine sulfate $\delta^{34}\text{S}$ reflecting oxygen-depleted and possibly also carbonate-oversaturated (i.e., alkalinity enriched) seawater during greater spreading rates and augmented MOR volcanism can be further supported by observed correlations between stratigraphically well-documented oceanic anoxic events (OAEs) and major volcanic episodes, accompanied also by enhanced formation of marine authigenic carbonates^{33,45}. Paleo-redox proxies and volcanic indices suggest an instantaneous response of ocean deoxygenation to submarine activity during the most severe Cretaceous OAEs^{47,48}, and Ca isotope data ($\delta^{44}\text{Ca}$) from OAE2 carbonate record also support such hypothetical increase in marine CaCO_3 saturation or alkalinity overshoot⁴⁵. Published data also show that the marine sulfate $\delta^{34}\text{S}$ record exhibits a strong positive shift at the onset of OAE2 (94.5 Ma), contemporaneous to the massive intra-oceanic, Caribbean–Colombian and Madagascar flood basalts^{47,48}, and by inference higher submarine volcanic activity with elevated inputs of reduced volatiles and gases in the ocean, leading to increased sediment pyrite burial under euxinic to anoxic conditions³³. In summary, our results align with the hypothesis of elevated sulfate $\delta^{34}\text{S}$ during ocean anoxia^{33,48} and suggest that the observed 60 Myr cyclicity in $\delta^{34}\text{S}$ proxy record, and potentially also marine carbonate chemistry (CaCO_3 cycle), is related to changes in the oceanic redox cycle and continental weathering rates both paced by global tectonics (Fig. 6).

60 Myr marine biogeochemical redox cycle paces biodiversity

The 60 Myr biodiversity cycle is in antiphase with the 60 Myr oceanic sulfate $\delta^{34}\text{S}$ and seafloor spreading rate cycles over the Phanerozoic (Supplementary Fig. S8 and Table S1), suggesting a potential connection between mid-ocean ridge dynamics and macroevolutionary processes (reflected in biodiversity) via marine redox cycling. Such anoxic or euxinic/sulfidic conditions, i.e., higher HS^- levels but lower dissolved O_2 in the ocean, would, in turn, negatively impact marine biodiversity³, as reflected in the coherent and strong anti-correlation between marine biodiversity and $\delta^{34}\text{S}$ record, as well as between extinction and $\delta^{34}\text{S}$ record (Supplementary Fig. S8 and Table S1). Geochemical models and observations clearly show such purported and significant secular variations in marine redox and O_2 levels in the ocean-atmosphere system during the Phanerozoic^{1,7,11,12,49,50}. Furthermore, local ocean anoxia or suboxic conditions, and accompanying deleterious impact on marine biota, have also been reported in recent MOR settings and spreading centers^{41,44,51}.

The record of a stronger 60 Myr signal in marine S (and Sr) isotope proxies during the Paleozoic versus the 28–38 Myr signals in tectono-eustatic data during the Ceno-Mesozoic (Fig. 4 and Supplementary Fig. S11, see also previous results in ref. 18.) may reveal different driving mechanisms and/or feedbacks on different timescales¹⁸. This does not mean that the driving mechanisms are decoupled at operating timescales, but one process or mechanism can dominate the other over different timescales. The shorter 28 and 38 Myr biodiversity cycles may be dominantly governed by tectonically driven sea-level change during the Ceno-Mesozoic¹⁸, while the longer 60 Myr biodiversity cycle reported and discussed here may result from long-term biogeochemical cycling and secular changes in paleo-seawater redox state, which may occur at a slower or longer pace^{12,29}. Although our results point to different driving mechanisms operating at different time scales

during the Ceno-Mesozoic vs Paleozoic intervals, it is also plausible that the temporal resolution and coverage of the available Paleozoic proxy datasets is not sufficient or dense enough to resolve the relatively shorter 28–38 Myr cycles. Importantly, focusing on the 60 Myr cycle band for secular variations in marine $\delta^{34}\text{S}$ record and the inferred paleo-seawater redox change (i.e., oxygenation vs. anoxia), its stronger amplitude during the Paleozoic matches the evolution of global tectonic activity (Fig. 2). This result points to more severe change in the amplitudes of Paleozoic ocean redox state that is in agreement with Earth system modeling showing stronger secular variations in atmospheric O_2 levels and thus marine redox during the Paleozoic (15 to 30%), compared to relatively weaker or muted variations (14 to 20%) during the Ceno-Mesozoic^{12,29}.

Hence, the link between MOR related processes and the notable changes in the amplitude of paleo-redox $\delta^{34}\text{S}$ proxy related cycle during the Paleozoic can be ascribed in part to the generally lower and gradually increasing levels of O_2 in the atmosphere-ocean system during the Paleozoic^{1,11,12}. Accordingly, Paleozoic seawaters were not fully saturated with respect to dissolved O_2 ¹, leading to higher sensitivity of the early ocean redox system, and by inference marine $\delta^{34}\text{S}$ proxy, to tectonically driven perturbations, the latter amplified by a relatively smaller marine sulfate reservoir (due to lower marine SO_4^{2-} concentrations) in the Paleozoic seawaters. Forced in turn by a stronger tectonic imprint during the Paleozoic, the marine redox system would thus strongly oscillate, fingerprinted in the prominent 60 Myr marine $\delta^{34}\text{S}$ proxy record and its fluctuations. Shallow-water marine habitats, which shelter majority of modern marine animals, and by analogy also housed the fossil counterparts, may have played a crucial role in reduction of biodiversity and enhancement of extinction process since these ecological environments are critically sensitive to redox cycling, especially in the relatively unstable atmosphere-ocean system of the Paleozoic¹. The potential connection between MOR process-driven deep-sea redox conditions and shallow-water habitats can be further augmented by ocean circulation that has also been demonstrated to be controlled by global tectonics⁷. The redox front, with oxygen-depleted and alkalinity-charged waters, can thus spread oceanwide to reach shallow marine or continental-shelf environments. These settings represent vulnerable areas sensitive to changes of physico-chemical and environmental parameters and marine biota susceptible to biological extinction and diversification⁵², while changes in marine carbonate chemistry (CaCO_3 saturation) influenced shallow-water biocalcification.

The macrostratigraphic carbonate dataset, which is an indicator of the extent of siliciclastic-starved and carbonate-rich shallow marine shelf environments, does not show a well-preserved 60 Myr cycle band. However, the greater amplitudes of this time series during the Paleozoic (Fig. 3) suggests that the influence of expanded epicontinental seaways on the biogeochemical cycles may have been significant. This may provide evidence for expansion and contraction of ecological niches for ancient calcifiers on shelves and in epeiric seas, under tectonically driven marine paleoredox cycling. Also, models and observations indicate the high sensitivity of shallow marine habitats and their contained biodiversity to changing O_2 levels¹⁴, and thus the observed shift in the amplitude of the 60 Myr carbonate depositional (CaCO_3) cycle from the mid-Permian to the latest Triassic can be related to changes in both ocean chemistry and marine carbonate producers. Specifically, dominant neritic or shallow-water calcifiers were present in the Paleozoic oceans, versus more pelagic organisms or open-ocean calcifiers (foraminifera, coccoliths) that controlled the marine carbonate cycle in the Ceno-Mesozoic oceans^{53,54}. The appearance of deep-sea pelagic calcifying organisms during the early Mesozoic may have thus buffered the severe changes in marine CaCO_3 cycle via the regulation of ocean carbonate factory (i.e., buffering 'alkalinity overshoot' events via pelagic carbonate formation)^{53,54}, leading to more balanced oceanic carbonate cycle and muted CaCO_3 fluctuations during the Ceno-Mesozoic, compared to the Paleozoic, as observed also in the 'carbonate area' data through time (Fig. 3).

The evolution of the 60 Myr $\delta^{34}\text{S}$ cycle also shows slightly elevated power at around 250 Ma (Fig. 2), which matches $p\text{O}_2$ maximum in previous

biogeochemical models⁵⁵. This would contradict with our hypothesis that deficiency of pO_2 hence instability of seawaters with respect to dissolved pO_2 would strengthen the amplitude of the 60 Myr $\delta^{34}S$ cycle. However, the Permian–Triassic transition at around 250 Ma was characterized by intensified global weathering⁵⁶ and ocean stratification⁵⁷ that may contribute in part to stronger amplitudes of the 60 Myr $\delta^{34}S$ cycle and ocean anoxia. Such scenario or interpretation is supported by the evolution of the 60 Myr cyclicity in the marine $^{87}Sr/^{86}Sr$ record, which clearly shows elevated powers at around 250 Ma (Fig. 2).

Collectively, our results and analysis of the Phanerozoic marine proxy data reveal a systematic coupling between global tectonic processes and biogeochemical cycling in the ocean. This tectonic driver controls the observed 60 Myr cycle in marine biodiversity via changes in continental weathering and seawater redox state, the latter impacting the environmental stress for marine biota and their extinction and diversity, especially within shallow marine habitats (Fig. 6). Our observations support recent Earth system modeling for the significant secular variations of Phanerozoic marine redox change and ocean oxygenation and its dependence on global plate tectonic configurations^{7,55}, lending support to arguments linking marine biotic crises and mass extinction events to oceanic redox state during the Phanerozoic^{2–4}.

The physical and chemical coupling between climate, seawater chemistry, and biogeochemical cycles through global tectonic forcing should be explored more systematically in the future, considering the complex evolution of solid Earth and its geodynamic history. Several geological episodes of intensified continental weathering and chemical alteration (i.e., water–rock interactions) occurred under an accelerated hydrological cycle and elevated paleotemperatures, suggesting that enhanced continental weathering may have occurred under more humid, warmer climatic conditions³³. Most Paleozoic extinction events are associated with climate warming and marine anoxia linked to fast global sea-level rises^{2,5,58,59}. The above processes may thus also explain or contribute to the observed strong coupling between the 60 Myr cycles in marine $^{87}Sr/^{86}Sr$ and $\delta^{34}S$ proxy records throughout the Phanerozoic, and particularly during the Paleozoic (Fig. 2). The three major mass extinction events, including the Late Ordovician, Late Devonian, end-Permian exhibit elevated marine $\delta^{34}S$ values, and their timings are close to the 60 Myr cycle maxima (Fig. 4).

Environmental stressors in marine settings, such as paleo-seawater redox conditions and O_2 and nutrient availability, along with other parameters such as temperature, pH, and sea level, can significantly impact shallow-marine or neritic environments in epicontinental/epiiric seas, thus governing the vulnerability of marine biota to extinction and diversification^{6,14,18}. Future studies should also focus on the potential role of marine carbonate system (alkalinity, pH, $CaCO_3$ saturation) and associated ocean acidification/alkalinization events constrained via B, Ca and Sr isotope proxies^{45,46,60–62}, as well as redox-driven processes (i.e., alkalinity produced via BSR - bacterial sulfate reduction) and their impact on marine biodiversity¹⁴ during the Phanerozoic. These goals can be achieved via the collection of new marine archives and high-resolution proxy records of past changes in seawater redox and carbonate chemistry, coupled with advanced Earth system modeling that will integrate geodynamics with global biogeochemical cycles.

Data and methods

Geological datasets

Fossil data were extracted from the Paleobiology Database (PBDB). We followed the procedure used in ref. 2 for consistency with past investigations of cyclicity. This procedure downloaded all metazoan (animal) fossil occurrences in collections tagged as from a marine setting (e.g., offshore, lagoonal). The final datasets (Supplementary Data Table 1) consisted of 1,043,231 fossil occurrences representing 41,457 genera after further filtering out occurrences not identified to the genus level or with Period-level or greater resolution. These culling procedures err on the side of data inclusivity, which has the benefit of producing a larger dataset but at the cost of potentially increasing the risk of Type II error from high taxonomic and/or

preservational variability. The fact that we observe 60 Myr cyclicity despite any such noise strengthens our conclusions. Furthermore, 60 Myr cyclicity is still observed in an alternative dataset limited to commonly-studied skeletonized marine invertebrates for greater preservational consistency (Supplementary Data Tables 2 and 3). All PBDB data are accessible via its API⁶³, which is updated continuously as new data are acquired. For convenience, and to allow precise reproduction of these results, concise raw data tables providing collection and taxon numbers, as well as PBDB ages used in this analysis, are provided in the supplementary materials (Supplementary Data Tables 1, 2 and 3). These data are also uploaded in the publicly accessible repository source DRYAD⁶⁴.

Time series analyses work best when there is consistent and even interval length among data points but most fossils are dated to geologic intervals of unequal durations⁶⁵. We addressed this problem by using a range-through methodology – i.e., inferring the presence of a taxon during an interval in which it is not observed based on its occurrence before and after ref. 18. Note that we serially “sample” the age timeline at arbitrary, but regular, increments (1 Myr) and determine the number of genera known to occur at each increment. The inherent temporal resolution of the fossil occurrence data are limited by the chronostratigraphic bins to which they have been assigned in the PBDB.

A variety of biodiversity time-series were calculated to summarize the PBDB data using the divDyn library¹⁰ for the R statistical software environment (R Core Team, 2024). The used data of divDyn calculations are provided in the supplementary materials (Supplementary Data Table 2), but the results presented here focus on the following: (1) range-through richness; (2) Shareholder Quorum Subsampling (SQS) richness⁶⁶; (3) per Capita extinction and origination⁶⁷; and (4) extinction and origination using the metric of Alroy⁶⁶, labeled “Alroy-2015”. These methods were applied in order to assess the stability of the 60 Myr cycle when using different biodiversity tabulation methods.

The marine macrostratigraphic carbonate and total marine sediment datasets for North America were extracted using the Macrostrat API⁶⁸ (see macrostrat.org/api). A total of 18,207 marine sedimentary rock units from 940 columns in North America were used to tabulate total marine sediment coverage area. Of these, 17,229 units are Phanerozoic in age. For convenience, and to allow precise reproduction of these results, a concise raw data table is provided in the supplementary materials (Supplementary Data Table 4).

A commonly cited concern with both fossil data from the Paleobiology Database and the Macrostrat database is that the disproportionate influence of North American data, or more generally systematic spatial heterogeneity in sampling, may distort results⁶⁹. We tested for this possibility by comparing North American to non-North American data in both the sediments and fossil datasets. The North American marine macrostratigraphy has been demonstrated to contain a globally-relevant signal²⁶, though it also contains the signature of regional tectonics unique to North America, but at higher orders⁷⁰. In particular, precursor and recent studies demonstrated the global nature of North America geological data at first and second orders including the 60 Myr cycle band we are exploring here in this study^{26,70,71}. We have further explored the global nature of North America stratigraphy by looking for a common 60 Myr cycle band in marine genus-level diversity in two PBDB datasets, one including only North America occurrences and the other considering all global occurrences except those from North America. Results show a prominent, common 60 Myr cycle band in both datasets (Supplementary Fig. S5), hence supporting the global nature of the 60 Myr cycle band. To complement the total marine sediment dataset we also calculated total carbonate-bearing marine sedimentary rock coverage. For this analysis, only those marine sedimentary rock units containing at least some amount of carbonate were used in the calculation. Alternatively, it is possible to restrict units on the basis of their estimated proportion of carbonate content. For the carbonate-dominant dataset, only those units estimated to be comprised of 50% or more carbonate were included in the calculation.

The tectonic data were inferred from the most recent global plate model⁷² that is updated from an earlier model⁷³, and extended to the whole

Phanerozoic Eon⁷⁴. This model additionally includes plate deformation along major rifts and orogens. It also considers the evolution of the age–area distribution of the ocean floor. This model provides plate subduction data and the length of the MOR, as well as the total variations in the crustal production rate through time. We have explored several tectonic parameter datasets, and have shown a strong 60 Myr signal in the mid-ocean spreading and subduction rates. We have also used deep Earth carbon (C) model data⁷⁵ to look for possible 60 Myr cycle band in tectonically derived C components. We have shown the 60 Myr cycle in global continental arc C outflux data, suggesting the potential impact of tectonic CO₂ degassing on the biogeochemical cycles. The biogeochemical proxy datasets include the marine carbonate strontium isotopes (⁸⁷Sr/⁸⁶Sr), which extends over the past ~506 Ma⁷⁶, and the marine sulfate sulfur isotopes (δ³⁴S), which extends over the past ~520 Ma⁷⁷. We have also used a more recent marine sulfate δ³⁴S dataset³⁰ to check the reliability of the 60 Myr cycle in the δ³⁴S record. All the ages of the analyzed geological datasets are updated according to the latest version of the time scales²⁰. We have explored the possible impact of shelf-area extent, at the studied 60 Myr cycle band, on the record of δ³⁴S related redox proxy. We have analyzed the residuals of δ³⁴S record for spectral analysis after removal of global shelf-area variations⁷⁸. Results indicate the persistence of the 60 Myr cycle band (Supplementary Fig. S12), hence concurring with the significance of δ³⁴S record for redox cycling at lower orders⁷⁷. Finally, we have explored the use of ⁸⁷Sr/⁸⁶Sr as a weathering proxy as follows. Radiogenic strontium isotope variations in seawater (ratios of ⁸⁷Sr/⁸⁶Sr) through time have been successfully deployed in chronostratigraphy (SIS – Strontium Isotope Stratigraphy) and have been also utilized as a proxy for intensity of weathering fluxes over the geological time scales of several Myr. The total input of Sr into the ocean at any given time, has two main origins or components, namely (i) the hydrothermal flux from the mantle or MOR systems, and (ii) the continental weathering flux dominated by more radiogenic Sr derived from felsic continental crust or lithosphere. There has also been an attempt to deploy the ⁸⁷Sr/⁸⁶Sr tracer as an index for changes in global rates of seafloor production and subduction^{79,80}, which is also a first order control on the long-term sea-level changes. It is assumed that if marine ⁸⁷Sr/⁸⁶Sr ratios could be corrected for the weathering input, then a sea-level component could be extracted. The weathering-component correction that van der Meer et al.⁷⁹ use is the global runoff (weathering) model⁸¹ that only includes the changes at the first-order time scale, i.e., ~250 Myr tectono-eustatic Wilson cycle scale^{35,82}, and which considers only the effects of changing paleogeography and solar flux. This approach would underestimate the 60-Myr ⁸⁷Sr/⁸⁶Sr cycle in the weathering component and overestimate it in the tectonic component. Recently, a ⁸⁷Sr/⁸⁶Sr proxy-based tectonic and sea level derived model⁸⁰ uses a new weathering-component model correction that includes paleogeography, solar flux and CO₂ degassing, but it is still at first-order scale⁸³. These corrections for the weathering component at only first order timescales^{79,80} would again overestimate the tectonic component at the 60 Myr cycle band in the marine ⁸⁷Sr/⁸⁶Sr record (Supplementary Fig. S13).

Thus, we see that the extraction of the tectonic component from the ⁸⁷Sr/⁸⁶Sr record should use a weathering model that considers changes at all time scales, and not only at the first order. Continental weathering should operate at different time scales and with different amplitudes driven by Earth's surficial and internal processes. For instance, at second order tectonic timescale, including the 60 Myr cycle, the weathering intensity should be modulated by mountain building and volcanic CO₂ degassing (balanced by subsequent uptake of CO₂ via enhanced silicate weathering). The tectonic data we are using here show prominent cyclic variations at tens of Myr scales (see also ref. 2). Also, the CO₂ signal, an important agent of continental weathering, which is documented in geological observations⁸⁴ shows strong variations at second order⁸⁵, while the used CO₂ degassing input in the weathering model is mostly at first order^{79,80}. Finally, the sea-level model data derived from the ⁸⁷Sr/⁸⁶Sr record shows strong 60 Myr fluctuations, while the relatively well-constrained Phanerozoic sea-level data from geological observations does not show the 60 Myr cyclicity^{18,82}. Even the resulting synthetic sea-level model curve that includes glacio-eustasy and

⁸⁷Sr/⁸⁶Sr derived tectono-eustatic component⁸⁰ does not correlate to the relatively well-constrained, observed Paleozoic sea-level data³⁶ (Supplementary Fig. S13). This points to potential difficulties in extraction of tectonic and weathering components from the ⁸⁷Sr/⁸⁶Sr record. We note that since continental weathering and tectonics are ultimately linked at the global scale, discrimination of each component from the ⁸⁷Sr/⁸⁶Sr record over the whole Phanerozoic Eon is hypothetical, and the 20% contribution of weathering component to the total ⁸⁷Sr/⁸⁶Sr variations⁸⁰ is unconstrained and may be too small (Supplementary Fig. S13). We thus use the ⁸⁷Sr/⁸⁶Sr fluctuations in marine carbonate record, in particular those at the 60-Myr cycle band, as a balance between the tectonic and weathering processes and their influence on the oceanic Sr isotope budget⁸⁶ (see below).

It should be noted that it has long been thought that fluctuations in seawater ⁸⁷Sr/⁸⁶Sr record during the Phanerozoic arise mainly from the two competing factors of (i) seafloor basalt alteration and hydrothermal venting from MOR systems promoting input of unradiogenic Sr into the ocean thus causing decrease of seawater ⁸⁷Sr/⁸⁶Sr, versus (ii) terrestrial weathering input supplied from continental interiors leading to more radiogenic Sr flux to the ocean and thus an increase in seawater ⁸⁷Sr/⁸⁶Sr^{87,88}. However, a more recent assessment of the modern oceanic Sr isotope budget indicates significant contributions (up to 60%) from the weathering of volcanic islands, island arcs, and oceanic islands (rather than from submarine MOR systems) supplying the unradiogenic Sr flux to the ocean via weathering, and the remaining ~40% of unradiogenic Sr is thus derived from seafloor basalt alteration and hydrothermal venting⁸⁶. Accordingly, these submarine or oceanic-derived non-radiogenic Sr contributions are balanced by radiogenic Sr inputs from continental weathering of more felsic crustal sources, and all these various inputs should be considered when interpreting the Phanerozoic seawater ⁸⁷Sr/⁸⁶Sr record in a global tectonic context^{86,89}. During collisions and subduction, mountain and continental arc building stimulates enhanced continental weathering due to higher topography or relief relative to sea level. In contrast, weathering of island arcs is inhibited because the arcs are submerged or buried in the suture zone, and thus, the balance between these two opposing effects tends to modulate and increase the seawater ⁸⁷Sr/⁸⁶Sr ratios during convergence⁸⁶. Accordingly, we suggest that the 60 Myr subduction rate cycle has modulated the intensity of weathering processes among continental and island arcs, leading to the observed 60 Myr ⁸⁷Sr/⁸⁶Sr cyclicity with shifts to more radiogenic values during times of increased subduction rates and thus plate convergence, and vice versa. It should be, however, also mentioned that the reconstructed global average convergence rates are different for the Paleozoic vs. Ceno-Mesozoic time intervals. Specifically, based on data and tectonic modeling⁷⁴ the mean crustal destruction rates along subduction zones are estimated at ~5.22 km²/yr for the Paleozoic compared to ~4.43 km²/yr for the Ceno-Mesozoic. Although the above difference (ca. 0.8 km²/yr) is not substantial it might have contributed to a possible shift in the balance of oceanic Sr fluxes during the Phanerozoic (i.e., from convergence-dominated to subduction-dominated systems), potentially causing the observed difference in response of marine ⁸⁷Sr/⁸⁶Sr proxy to global tectonic (plate convergence) forcings, making the 60 Ma cycle more prominent in the Paleozoic compared to the Mesozoic/Cenozoic (Fig. 4).

A number of recent studies have also shown the predominant role of convergent margins in global weathering fluxes and in controlling the oceanic Sr elemental and isotope budget^{90–93}. In particular, it has been postulated that the production of thick and extensive continental arcs and crust during active plate subduction would generate continental elevation (mountain building), stimulating the weathering and erosion processes leading to higher proportions of radiogenic Sr input in the ocean, increasing marine ⁸⁷Sr/⁸⁶Sr^{86,93–95}. It has also been suggested that fluctuations in Phanerozoic seawater ⁸⁷Sr/⁸⁶Sr record do not reflect the temporal changes in global chemical/silicate weathering rate but rather changes in the Sr isotopic composition of the primary continental sources, such as silicic magmatism, i.e., the relative proportion of more evolved versus less evolved magmas in the continental crust exposed to weathering^{89,90}. Our results therefore confirm the in-phase relationship between the 60 Myr global subduction and

marine $^{87}\text{Sr}/^{86}\text{Sr}$ cycles, particularly throughout the Paleozoic, hence supporting the idea of the critical role of convergent margins in driving the ocean chemistry and marine Sr isotope budget through geological time⁹³, but does not rule out the hypothesis that paleo-seawater $^{87}\text{Sr}/^{86}\text{Sr}$ record reflects the magmatic evolution of the weathered continental crust sources⁹⁰.

Time-series analysis methods

Long-term irregular trends in the geological datasets were measured and removed using the lowess weighted average⁹⁶ and/or the polynomial method. The goal of the use of different methods of detrending is to test the effect of the detrend on the spectral detection of the targeted cycles (e.g., Supplementary Fig. S1). For spectral analysis, we used the multi-taper spectral method (MTM) associated with the harmonic F-test, with three 2π prolate tapers⁹⁷. The F-test was used to seek evidence for individual lines within frequency bands with elevated amplitude that could be related to specific significant cyclicalities. The 2π -MTM amplitude spectra are compared to the basic Fourier transform squared Periodogram (unsmoothed periodogram, Matlab's periodogram.m) along with the classical autoregressive AR(1) red noise model. The F-test is obtained after spectral background estimation based on the LOWSPEC method⁹⁸. Spectral analysis is applied to the whole Phanerozoic geological datasets, then per intervals namely the Ceno-Mesozoic and Paleozoic intervals with a focus on the Paleozoic because the 60 Myr signal is stronger. Nevertheless, to amply recover the 60 Myr cycle band the two intervals 0–300 Ma and 200–540 Ma are privileged (Supplementary Figs. S1 and S2). Note that $\delta^{34}\text{S}$ data (ref. 75) don't cover all the Phanerozoic (see 'Geological datasets'). Spectral analysis outputs of these extended time intervals provide more precision on the determination of the periodicity of the 60 Myr cycle. We tested the statistical significance of the 60 Myr cycle band after Bonferroni correction for multiple frequencies (Supplementary Fig. S1). We also considered the spectra for a low-order detrend of the data to reduce increased rates of false spectral detection at low frequencies (Supplementary Fig. S1). We equally applied the Lomb-Scargle (L-S) periodogram to the low-order detrended biodiversity data and showed the persistence of the significance of the 60 Myr cycle (Supplementary Fig. S1). To further check the validity of the significant spectral peak related to the 60 Myr cycle, identified in the standard power spectra, we additionally used the Bayesian probability spectra⁹⁹ (Fig. 1).

We used the Gaussian filter to bandpass filter the targeted cycles¹⁰⁰ paired with the singular spectrum analysis (SSA)¹⁰¹. We applied the SSA as an independent technique to FFT based methods, which quantifies both cyclic and acyclic components in a signal without forcing a periodic background to the analyzed time series. For instance, the tectonic data show low-frequency (cyclic and acyclic) variations with stronger amplitudes. To remove these strong tectonic variations (≥ 100 Myr) we used the SSA method, with a 100 Myr embedding dimension and the three first reconstructed components (RC) were removed in order to highlight high-frequency cyclicalities (≤ 6 Myr) (Fig. 5 and Supplementary Figs. S10 and S14). Finally, correlation among the analyzed geological cyclicalities was quantified using the 2π -MTM cross-spectral analysis provided in the Matlab routine¹⁰² (Supplementary Figs. S7 and S8).

Because the cyclic nature of the fossil records is debated, we applied several methods to SQS biodiversity data. A previous study used classical FFT based periodogram along with Monte Carlo simulations on the power spectrum of biodiversity data to show a high significance of the 60 Myr spectral peak. Here, in addition to our use of various spectral analysis methods, the MTM associated with the F-test, the classical FFT periodogram associated with the classical red noise model, the L-S periodogram, which all of them show a significant 60 Myr cyclicity in the biodiversity patterns, we run direct Monte Carlo simulations on SQS diversity data assuming their statistical properties (e.g., variance) in either a gaussian fit or a mixed gaussian-exponential fit¹⁰³. We found a high significance of the 60 Myr cyclicity (Supplementary Fig. S15), hence supporting previous studies for the statistical significance of the 60 Myr biodiversity cycle^{17,18,104}. The Bayesian probability spectra are employed as an independent partial check of the validity of the significant 60 Myr spectral peak identified in the

standard power spectra⁹⁹. Result provides confirmation of a significant 60 Myr frequency sinusoidal component in the biodiversity data (Fig. 1). Finally, the SSA method¹⁰¹ (see above) is applied to the SQS biodiversity data as another independent partial check of results obtained by FFT based methods. Result highlights the 60 Myr cyclicity superimposed on long-term biodiversity trends (Supplementary Fig. S15).

We studied the evolution of the power of the analyzed, targeted cyclicalities in geological proxy datasets throughout the Phanerozoic Eon using the Matlab script pda.m¹⁰⁵. The pda.m function, which is based on 2π -MTM power spectra, measures the ratio of the power of a targeted cyclicity (e.g., 60 Myr cycle band) over the total power (0 to Nyquist band). We restricted the analyses from 0 to 0.1 cycles/Myr for biodiversity, biogeochemical and tectonic datasets, and from 0 to 0.05 cycles/Myr for extinction and origination datasets because the stronger high-frequency variabilities in these data. We also removed long-term variations prior to pda.m estimates. In particular, the tectonic data are characterized with strong low-frequency (≥ 100 Myr) variations that we removed to highlight cycles ≤ 6 Myr frequency bands. Irregular variations and trends in the biodiversity data are also removed prior to pda.m in a similar fashion as for spectral analysis (i.e., 50% wa for SQS and 35% wa for RT). The aim of analyzing the evolution of the recorded cyclicalities over the Phanerozoic is to assess whether the Ceno-Mesozoic and Paleozoic intervals have different behavior in terms of the expression of cyclicalities¹⁸, hence can potentially be paced by different forcing mechanisms.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The paleobiodiversity datasets used in this study are available at the below DRYAD link: http://datadryad.org/share/iXJ6ma8EELtJN00vG1wQ4eLDTdd6JgL8_ev7vdasQI

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Author contributions

S.B. designed the study, and performed time-series analysis. R.D.M. contribute by the significance and interpretations of the tectonic data. S.E.P. and A.Z. contributed by the biodiversity and macrostratigraphic datasets and the generation of multiple fossil datasets using different metrics and approaches, as well as the significances and environmental interpretations of biodiversity. J.F. contributed by interpretations of geochemical datasets. BUH contributed by interpretations of all geological datasets. S.B., S.E.P., R.D.M., J.F., and B.U.H. discussed together the interpretations. S.B. wrote the manuscript with contributions from S.E.P., R.D.M., J.F. and B.U.H. All authors edited the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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